Solving Chess Endgames

A Reinforcement Learning Approach

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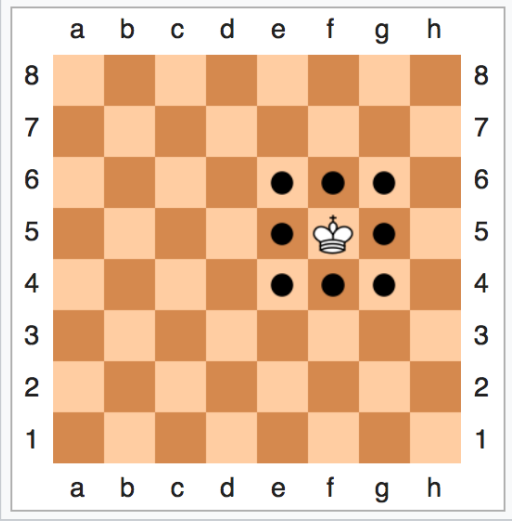
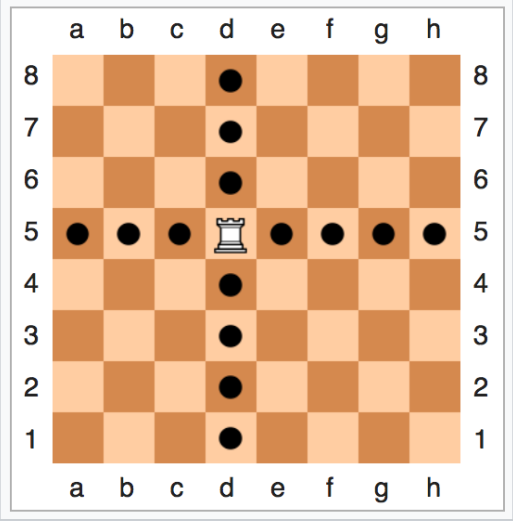
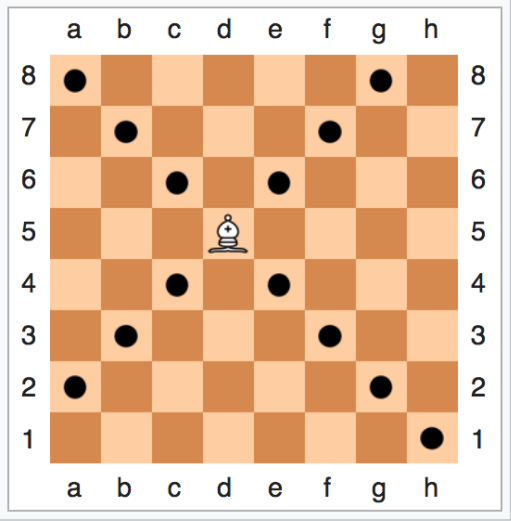
Chess Programming Background

- 1890 El Ajedrecista: electro-mechanical KRK solver
- 1950 Claude Shannon: first article about chess programming
- 1997 *Deep Blue*: first computer program to beat the current world champion
- 2008 *Stockfish*: strong heuristic-based chess engine
- 2017 *Alphazero*: neural network-based approach that defeated Stockfish
- from 2018: more models developed based on *Alphazero*

Chess Rules

- **Win**: checkmate the opponent's king
- **Draw**: stalemate or insufficient material[1]

[1] threefold repetition and the fifty-move rule are not considered in our implementation

Elementary Endgames

- subset of endgames with few pieces left
- clear theoretical results: forced checkmate or draw
- foundation for learning more complex endgames
- example: KRK
- solved with tablebases up to 7 pieces

Remark: white is always winning in elementary endgames!

MDP Formulation

Markov Game vs MDP

- Chess is a 2-player, turn-based, zero-sum game, formally a Markov Game
- Assume fixed black defender policy: **oracle** or **random**
- Black becomes part of the environment: Markov Game $\rightarrow$ MDP

MDP: States

- State space $\mathcal{S}$: legal pieces positions and side-to-move

```
8/8/K7/8/8/5k2/7R/8 w - - 0 1
```

```
std::array<std::array<U64, 6>, 2> pieces;
Color side_to_move;
```

- Terminal state space $\bar{\mathcal{S}} \subset \mathcal{S}$: checkmate or draw positions (and stm). Defined procedurally

```
state.is_game_over() -> bool
```

MDP: Actions

- Action space $\mathcal{A}(s)$: set of legal moves for the current player.
- Defined procedurally:

```
state.legal_moves() -> std::vector<Move>
```

MDP: Transition

Transition probability $\mathbf{Pr}(S_{t+1} = s' | S_t = s, A_t = a)$

- **Oracle Defender**: deterministic evolution

```
state.do_move(Move)                 // white deterministic
state.do_move(oracle.reply())       // black deterministic
```

- **Random Defender**: stochastic evolution

```
state.do_move(Move)                   // white deterministic
state.do_move(random_reply())         // black random
```

MDP: Reward

Goal: find the fastest mate

Reward:[2]

- checkmate: +1
- draw: -1 (always winning for white)
- encourage faster mates
  - step penalty: -0.01
  - discount factor $\gamma = 0.99$

[2] Except for Value Iteration

RL Algorithms

Monte Carlo Tree Search

- One of the oldest methods to solve chess
- Explore the game tree by simulating moves

Dynamic Programming

Value Iteration, for estimating $\pi \approx \pi_*$

Algorithm parameter: a small threshold $\theta > 0$ determining accuracy of estimation
Initialize $V(s)$, for all $s \in \mathcal{S}^+$, arbitrarily except that $V(terminal) = 0$

Loop:
| $\Delta \leftarrow 0$
| Loop for each $s \in \mathcal{S}$:
| $v \leftarrow V(s)$
| $V(s) \leftarrow \max_a \sum_{s',r} p(s', r|s, a)\big[r + \gamma V(s')\big]$
| $\Delta \leftarrow \max(\Delta, |v - V(s)|)$
until $\Delta < \theta$

Output a deterministic policy, $\pi \approx \pi_*$, such that
$\pi(s) = \text{argmax}_a \sum_{s',r} p(s', r|s, a)\big[r + \gamma V(s')\big]$

- KRk: 182.676 states
- KQk: 152.968 states
- KBBk: ? states

Choice of parameters

- Step penalty: -2 $\rightarrow$ values = DTZ
- Draw penalty: -100 $\rightarrow$ slow mates > quick draws
- $\gamma = 1$
- $\theta = 0.0001$

Random Player

Strong assumption: opponent plays optimally! Idea: train against an opponent that plays at random and see how well it can play against an optimal one.

Q-Learning: model-free approach

Assume no information on transitions!

Q-learning (off-policy TD control) for estimating $\pi \approx \pi_*$

Algorithm parameters: step size $\alpha \in (0, 1]$, small $\varepsilon > 0$
Initialize $Q(s, a)$, for all $s \in \mathcal{S}^+, a \in \mathcal{A}(s)$, arbitrarily except that $Q(terminal, \cdot) = 0$

Loop for each episode:
  Initialize S
  Loop for each step of episode:
    Choose A from S using policy derived from Q (e.g., ε-greedy)
    Take action A, observe R, S'
    $Q(S, A) \leftarrow Q(S, A) + \alpha\big[R + \gamma \max_a Q(S', a) - Q(S, A)\big]$
    $S \leftarrow S'$
  until S is terminal

Off-policy: we don't care about the behavior policy!

Choice of parameters

- Max steps per episode: 50 $\rightarrow$ if it's too far from mate, agent may never reach terminal state
- Step penalty: 0 $\rightarrow$ otherwise would learn wrong values from early termination
- Draw penalty: -1
- $\gamma = 0.99$
- Learning rate $\alpha = 0.05$
- ϵ-greedy policy with $\epsilon = 0.3$ with decay

Full Reinforcement Learning

- Problem: too many states, **curse of dimensionality**!
- Idea: use neural networks for policy and linear approximation for value function

One-step Actor–Critic (episodic), for estimating $\pi_{\boldsymbol{\theta}} \approx \pi_*$

Input: a differentiable policy parameterization $\pi(a|s,\boldsymbol{\theta})$
Input: a differentiable state-value function parameterization $\hat{v}(s,\mathbf{w})$
Parameters: step sizes $\alpha^{\boldsymbol{\theta}} > 0$, $\alpha^{\mathbf{w}} > 0$
Initialize policy parameter $\boldsymbol{\theta} \in \mathbb{R}^{d'}$ and state-value weights $\mathbf{w} \in \mathbb{R}^{d}$ (e.g., to $\mathbf{0}$)
Loop (for each episode):
  Initialize S (first state of episode)

  Loop while S is not terminal (for each time step):
    $A \sim \pi(\cdot|S,\boldsymbol{\theta})$
    Take action A, observe S', R
    $\delta \leftarrow R + \gamma \hat{v}(S',\mathbf{w}) - \hat{v}(S,\mathbf{w})$ (if S' is terminal, then $\hat{v}(S',\mathbf{w}) \doteq 0$)
    $\mathbf{w} \leftarrow \mathbf{w} + \alpha^{\mathbf{w}} \delta \nabla \hat{v}(S,\mathbf{w})$
    $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \alpha^{\boldsymbol{\theta}} \delta \nabla \ln \pi(A|S,\boldsymbol{\theta})$
    $S \leftarrow S'$

Approximations

- Policy function approximation: neural network
- State space approximation: linear combination of features
- Feature engineering:
  - Manhattan distance between kings
  - Manhattan distance between black king and white rook
  - File and rank of the kings
  - Distance of the Black King from the side of the board
  - Distance of the black king from the corner of the board
  - Binary indicator of opposition

Results

Assessment Metrics Overview

* DTM, DTZ etc * success, top1 etc

Why no convergence?

- Difficulties:
  - too many possible actions at each step
  - ideal opponent that never makes mistakes
  - very sparse rewards in an episode
- Possible solutions:
  - more training time
  - implement draw by repetition and 50-move rule $\rightarrow$ more frequent terminal states

Policy interpretability through human chess principles

Video Animation of optimal ply vs our policy

maybe for a mate in 5 or more

Future Works

* zobrist hashing and invariance properties in chess endgames to reduce state space cardinality

Why is it so difficult to solve chess endgames?

> *A player can sometimes afford the luxury of an inaccurate move, or even a definite error, in the opening or middlegame without necessarily obtaining a lost position. In the endgame ... an error can be decisive, and we are rarely presented with a second chance.*

- Paul Keres

Thank You!

Why $\gamma = 0.99$?

The longest rkr endgame requires 16 white moves, hence $\gamma^{16} \approx 0.85$ is a good reward for reaching the goal and producing fast mates.